

CHARAN YAKASIRI

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PROFESSIONAL SUMMARY

Aspiring *FPGA and Low-Level Systems Engineer* focused on building efficient digital systems and growing across RTL, system-level architecture, and chip design.

EDUCATION

B.E – Electronics and Communication Engineering

Saveetha Engineering College, Chennai

JUN 2023 - MAY 2027

GPA: 8.0

Higher Secondary Education – PCM (Class XII)

Terapanth Jain Vidyalaya Mat. Hr. Sec. School, Chennai

JUN 2021 - MAR 2023

GPA: 8.5

SKILLS

Core Skills: RTL Design, FPGA Synthesis & Implementation, FSM Design, IP Integration, Low-Level System Design

Languages: Verilog HDL, SystemVerilog, C, Python, TCL

Technologies: FPGA, RTL to GDS II, RISC-V ISA, CDC, Hardware-Software co-design, Timing Analysis, RTOS

Tools: Xilinx Vivado, Intel Quartus, ModelSim, Yosys, nextpnr, OpenROAD, QNX Momentics IDE, Cadence Virtuoso

Boards: PYNQ-Z2, Spartan-7, VSDSquadron FM, Vicharak Shrike Lite

Other: Linux, Git, Shell Scripting

PROJECTS

FPGA-Based Multi-Camera Aggregator IP Core (PYNQ-Z2) | [GitHub](#)

- Architected a *dual-camera video aggregation pipeline* in PL fabric; designed 7 *custom RTL IP's* including a Sync Unit, Arbitration FSM, Frame Alignment Buffer, and AXI-Lite Register Bank.
- Managed *CDC* between a *74.25 MHz HDMI clock* and *100 MHz AXI system clock* using an asynchronous FIFO with Gray-code pointer synchronization (<0.1% frame loss); executed its complete RTL-to-GDSII flow using Cadence Genus and Innovus.
- Eliminated AXI crossbar address collisions by migrating to *AXI SmartConnect for parallel HP port binding*, enabling runtime frame-boundary-safe camera switching via PS over AXI-Lite MMIO.

RTL-Based CNN Accelerator on FPGA (PYNQ-Z2) | [GitHub](#)

- Led *HW-SW co-integration and custom IP packaging* for a CNN accelerator, leveraging AXI-Stream and AXI-Lite interfaces to achieve an optimized *0.89 ms system latency* and *1121 FPS throughput*.
- Developed Python-based image preprocessing and postprocessing pipelines, establishing an automated validation framework to verify *PS-PL data transfer integrity* over *1000+ executions*.

Custom IP Integration on RISC-V (FPGA) | [GitHub](#)

- Optimized *GPIO & PWM IPs* in Verilog, achieving less than *5% LUT utilization* and *sub-10ns latency* via memory-mapped RISC-V integration.
- Developed *2KB bare-metal C firmware* to validate hardware, achieving *100% test coverage* across PWM frequencies using Yosys / IceStorm on VSDSquadron FM.

PROFESSIONAL EXPERIENCE

Combat Vehicle Research and Development Establishment

Intern (EMI/EMC)

Chennai, India

JAN 2025

- Executed 8+ EMI/EMC tests using spectrum analyzers; maintained 20+ compliant test records for automotive and defense systems.
- Evaluated noise immunity across multiple scenarios, applying shielding techniques to reduce interference by 20–30%.

CERTIFICATIONS & ACHIEVEMENTS

- DVCON (2nd Round)** – My work is on Semantic Gating Arch on Custom built RISC-V and RTL logic (in progress)
- Samsung ISWDP Fellow (Cohort 4)** – Gained exposure to semiconductor manufacturing and Synopsys TCAD tools
- FPGA Design Program – VSD** – Built and integrated custom IPs (GPIO, PWM) with RISC-V on VSDSquadron FM
- RTOS using QNX – Pi Square Technologies** – Studied QNX microkernel architecture using Momentics IDE
- Contributed to NSS initiatives by *teaching and improving rural school infrastructure*, enhancing learning conditions
- Organized and hosted *school and college events*, managing execution and strengthening leadership and communications